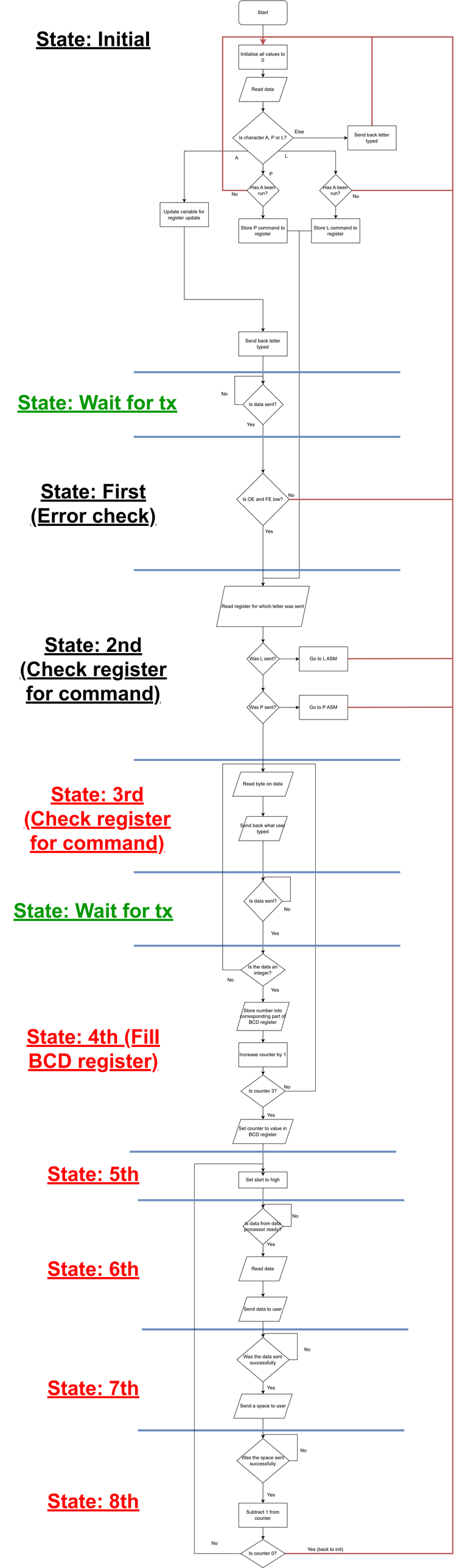
State: Initial
Start
Initialise all values to 0
Read data
Is character A, P or L?
Else
Send back letter typed
A
L
P
Has A been run?
No
Has A been run?
No
Update variable for register update
Store P command to register
Store L command to register
Send back letter typed
State: Wait for tx
No
Is data sent?
Yes
State: First (Error check)
Is OE and FE low?
No
Yes
Read register for which letter was sent
State: 2nd (Check register for command)
Was L sent?
Go to L ASM
Was P sent?
Go to P ASM
State: 3rd (Check register for command)
Read byte on data
Send back what user typed
Is data sent?
No
State: Wait for tx
Yes
Is the data an integer?
No
Yes
Store number into corresponding part of BCD register
State: 4th (Fill BCD register)
Increase counter by 1
Is counter 3?
No
Yes
Set counter to value in BCD register
State: 5th
Set start to high
Is data from data processor ready?
No
Yes
State: 6th
Read data
Send data to user
Was the data sent successfully
No
Yes
State: 7th
Send a space to user
Was the space sent successfully
No
Yes
State: 8th
Subtract 1 from counter
Is counter 0?
No
Yes (back to init)